

Affan Arif Khamse

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Education

Master of Science in Computer Science (GPA: 3.8/4.0)

New York University

New York, USA

May 2025

Bachelor of Engineering in Computer Engineering (GPA: 9.00/10.0)

University of Mumbai

Mumbai, India

June 2023

Technical Skills

Programming and Scripting Languages: Java, Python, TypeScript, JavaScript, C++

Frameworks and Libraries: Spring Boot, FastAPI, Flask, Django, Next.js, React, LangChain, Node.js

Cloud & Infrastructure: AWS (EC2, DynamoDB, S3, Lambda, SQS, SES), GCP, Docker, Kubernetes

Databases: SQL (MySQL, PostgreSQL), NoSQL (MongoDB, DynamoDB), Supabase, Redis, Pinecone

Testing & Monitoring: PostHog, JUnit, PyTest, Django Test, CloudWatch, Chrome DevTools

DevOps & CI/CD: GitHub Actions, Travis CI, Docker Compose, AWS SAM, Git, Swagger/OpenAPI

System Design: Agentic Systems, Microservices, Event-Driven, Distributed Systems, RESTful APIs

Work Experience

AI Engineer

New York, USA

Centh AI (South Park Commons-backed)

May 2026 – Present

- Forward-deployed across enterprise clients, owning system design end-to-end by translating live client requirements directly into production build decisions across concurrent client conversations
- Engineered an agentic credentialing pipeline from scratch, a database-backed queue orchestrating parallel verification agents (Claude Haiku for LLM checks, Mistral for OCR), processing **800+ documents/day** in production with automated recruiter handoff
- Shipped a candidate-facing **engagement agent** coordinating clinical test scheduling and appointment logistics (e.g., TB screenings) ahead of onboarding completion, reducing manual back-and-forth between candidates and recruiters
- Extended a **500+ emails/day** screening agent with a human-feedback loop updating agent prompts from recruiter corrections; leading consolidation of two production codebases (GCP Python backend and Next.js TypeScript platform) into a unified architecture

AI Software Engineer

New York, USA

Novum AI

Jun 2025 – Apr 2026

- Architected an AI call assistant end-to-end, reducing suggestion latency from **3.4s to under 1.5s** via parallelized retrieval and 72% prompt token reduction (4,900 to 1,950)
- Redesigned the retrieval architecture with **layout-aware hierarchical chunking** and **hybrid search** combining dense embeddings with sparse BM25 vectors, replacing flat chunking that destroyed document structure
- Deployed **event-driven backend services** with AWS SAM, Redis-backed session auth, **multi-tenant RBAC**, and production security hardening (webhook verification, CORS enforcement, injection prevention)

Projects

Stoca - AI-Native Local Commerce Platform

April 2026 - Present

Next.js, TypeScript, FastAPI, Python, PostgreSQL, Supabase, pgvector, Claude API, GitHub Actions

[GitHub](#)

- Built an AI-native commerce platform where store owners manage their business through conversation with a Claude-powered assistant supporting **19 tool calls** via Vercel AI SDK streaming, covering inventory, pricing, orders, and promotions
- Integrated **Claude Vision**, **pgvector semantic search**, and Pexels API into an autonomous product enrichment pipeline for shelf-photo identification and catalogue matching

FlashBids - Anti-Sniping Auction Platform

Sep 2024 - Dec 2024

AWS EC2, DynamoDB, S3, Redis, WebSockets, CloudWatch, Flask, Python, REST APIs, DevOps

[Demo](#) | [GitHub](#)

- Created a **real-time auction platform** using Flask and WebSockets with automatic time extensions to prevent last-second bid sniping, deployed on AWS EC2 with Auto Scaling and DynamoDB
- Profiled Redis connection bottleneck (50-80ms per new connection during bid floods) and implemented connection pooling, dropping **p95 latency from 800ms to 200ms**

Leadership & Technical Contributions

- **Mentored 50+ NYU students** in C++ and Java as Teaching Assistant across 4 semesters; several secured internships post-course
- Introduced company-wide **Ruff / ESLint / Prettier** setup at Novum AI, improving code consistency and reducing review cycles